# ISAAC ROSENBERG

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## Education

- **Erasmus Mundus Joint Master in Astrophysics** *September 2025 – July 2027*
  *Master of Science in Astrophysics* Italy, Serbia, France
  - Two-year joint program across University of Rome Tor Vergata, University of Belgrade and Université Côte d'Azur focused on astrophysics and space science.
  - **Current grade:** 29.7 / 30

- **University of Toronto** *September 2021 – May 2025*
  *Honours Bachelor's of Science* Toronto, Canada
  - Astronomy and Astrophysics Specialist, Mathematics Minor.
  - **Grade:** 3.85 / 4.00

## Publications

1. **Isaac Rosenberg**, Martine Lokken, Renée Hložek, Weiguang Cui, Sara Santoni, Romeel Davé (2026). **Here, There and Everywhere: How AGN Jets Affect Galaxy Cluster Environments**. Submitted for publication in *Astronomy & Astrophysics*.

## Research Experience

**Weak Lensing with SKA** *2026 - 2027*
*Observatoire de la Côte d'Azur, Université Côte d'Azur*
- Master's thesis project with Professor André Ferrari and Professor Simon Prunet.
- Building ML pipelines to measure cosmic shear with SKA.

**The Impacts of AGN Feedback on Galaxy Cluster Environments** *2024 - 2026*
*Department of Astronomy, University of Toronto*
- Research project with Professor Renée Hložek and Dr. Martine Lokken.
- Ran hydrodynamic simulations with the 300 collaboration of galaxy clusters to understand how AGN feedback affects galaxy and cluster properties.
- Paper submitted to Astronomy & Astrophysics.

**Standard Siren Cosmology with Gravitational Waves** *2024 - 2026*
*Canadian Institute for Theoretical Astrophysics*
- Research project with Professor Maya Fishbach.
- Built a model that can simultaneously fit for the cosmological and astrophysical evolution of the binary black hole mass distribution.
- Received Summer Undergraduate Research Fellowship to continue this project over summer 2025.
- Paper in preparation.

**Summer Research, ATLAS Experiment** *Summer 2024*
*CERN, Switzerland*
- Worked under Professors Nikolina Ilic and Richard Teuscher on the ATLAS experiment at CERN for 12 weeks, in person.
- Investigated yearly spatial variations in jet distributions of the $H \to ZZ$ particle decay.

## Conferences, Schools & Talks

1. **The Three Hundred Annual Collaboration Meeting** *August 2026*
   *La Cristalera, Spain* Talk
   Research talk on AGN feedback simulations; five days of discussions and workshops on galaxy cluster simulations.

2. **International Conference on Physics Students** *August 2026*
   *Rabat, Morocco* Talk
   Research talk on gravitational wave cosmology; five days of lectures and workshops across broad topics in physics.

3. **Flatiron Institute CMB Summer School** *July 2026*
   *New York, US* School
   Five-day workshop on CMB data analysis with the Simons Observatory.

4. **Master in Astrophysics and Space Science Winter School** — *Feb 2026*
   *Rome, Italy* — School
   Four-day workshop on machine learning and data science applications to astrophysics.

5. **FLATs Talk Series** — *January 2026*
   *University College London, UK* — Talk
   Research talk on galaxy clusters.

6. **Cosmology Meeting** — *January 2026*
   *Imperial College London, UK* — Seminar
   Research talk on galaxy clusters.

7. **Simulations Meeting** — *January 2026*
   *University of Oxford, UK* — Seminar
   Research talk on galaxy clusters.

8. **CMB – Large Scale Structure Meeting** — *January 2026*
   *University of Cambridge, UK* — Seminar
   Research talk on galaxy clusters.

9. **EXPLORE Summer School** — *August 2025*
   *Toronto, Canada* — Talk
   Research talk on gravitational wave cosmology; winner of best talk award.

10. **CASCA Conference** — *June 2025*
    *Halifax, Canada* — Poster
    Research poster and talk on galaxy clusters.

11. **STATSTRO Conference** — *May 2025*
    *Toronto, Canada* — Talk
    Research talk on gravitational wave cosmology.

## Honors and Awards

- **Winner of Best Talk Award at EXPLORE Summer School** — *2025*
  *York University*
- **Erasmus Mundus Joint Masters Scholarship** — *2025 – 2027*
  *Erasmus MASS program* — €34,000
- **Summer Undergraduate Research Fellowship** — *2025*
  *Canadian Institute for Theoretical Astrophysics* — $9,500
  - Summer fellowship to research gravitational wave cosmology.
- **CIE International Experience Award** — *2024*
  *Centre for International Experience, University of Toronto* — $1,500
- **Helen Mae Woodliffe Scholarship** — *2024*
  *Victoria College, University of Toronto* — $1,000
- **John Hamilton Watson Award for International Study** — *2024*
  *Victoria College, University of Toronto* — $3,000
- **Prof. William Kingston and Dr. John Kingston Scholarship** — *2023*
  *Victoria College, University of Toronto* — $1,000
- **Regents In-course Scholarship I** — *2022*
  *Victoria College, University of Toronto* — $1,000
- **Regents Admission Scholarship** — *2021*
  *Victoria College, University of Toronto* — $5,000

## Teaching Experience

**Teaching Assistant** — *2023 - 2025*
*University of Toronto, Department of Mathematics & Department of Astronomy*

- Teaching assistant for MAT186 (Calculus I), MAT187 (Calculus II), MAT188 (Linear Algebra), and AST251 (Life on Other Worlds).
- Led three tutorials per week, marked assignments, invigilated exams, and held office hours for university calculus, linear algebra, and astronomy courses with over 4,000 students combined.

## Relevant Coursework

** Graduate-level courses in italics*

**Astrophysics**

- **Courses:** Stars and Planets, Galaxies and Cosmology, Astrophysics Laboratory, *Cosmology*, *Stellar Astrophysics*, *Radiative Processes*, *Active Galactic Nuclei*, *Plasma Spectroscopy*, *Computational Astrobiology*, *Interstellar Medium*, *General Relativity*, *Formation of Galaxies*, *Astronomical Software Engineering*.
- **Textbooks:** Carroll & Ostlie (*An Introduction to Modern Astrophysics*); Ryden (*Introduction to Cosmology*); Sparke & Gallagher (*Galaxies in the Universe*); Baumann (*Cosmology*, Ch. 1–7); Lequeux (*The Interstellar Medium*, Ch. 1–12); Weinberg (*Gravitation and Cosmology*); Rybicki & Lightman (*Radiative Processes in Astrophysics*, Ch. 1–7).

**Physics**

- **Courses:** Classical Mechanics I & II, Electromagnetism I & II, Thermal Physics, Computational Physics, Physics Laboratory, *High Energy Physics*, *Quantum Mechanics*.
- **Textbooks:** Griffiths (*Introduction to Electrodynamics*); Morin (*Introduction to Classical Mechanics*, Ch. 1–7); Landau & Lifshitz (*Mechanics*); Cohen-Tannoudji, Diu & Laloë (*Quantum Mechanics Vol. I*); McIntyre (*Quantum Mechanics*); Newman (*Computational Physics*); Griffiths (*Introduction to Elementary Particles*, Ch. 1–4, 6–9, 11); Schroeder (*An Introduction to Thermal Physics*, Ch. 1–7).

**Mathematics and Computer Science**

- Analysis, Multivariable Calculus, Linear Algebra I & II, Differential Equations, Introduction to Computer Programming, Neural Networks, *Complex Analysis*, *ODEs & PDEs*.

## Skills

- **Programming Languages:** Python, C, C++, Perl, Java, MATLAB, ROOT
- **Hydrodynamic Simulations (and related tools):** GIZMO, GADGET, CAESAR, YT, DisPerSE
- **Python Packages:** Numpy, Scipy, Matplot, Pandas, Astropy, PyTorch, scikit-learn

## Professional Memberships

- **The Three Hundred Collaboration** *October, 2024 - Present*

## Music

- **Vocalist, Yiddish Glory** *2016 - Present*
  - Lead vocalist on the albums *Yiddish Glory: The Lost Songs of World War II* and *Yiddish Glory: The Silenced Songs of the Soviet Yiddish Theatre*.
  - Grammy Award nominee (Best World Music Album 2019) and recipient of the German Critics Prize (Best Ethnic Album, 3rd Quarter 2026).
  - Performed dozens of concerts worldwide, including in Canada, Germany, Austria, and Czechia.
  - Sang newly discovered music written by Holocaust survivors and victims in the Soviet Union.

## Additional Information

**Languages:** English (native), French (intermediate), Russian (intermediate), German (beginner)
**Interests:** Guitar, singing, world history, baseball, travel, chess, politics, music, football.